

LinuxTM



Linux

- ▶ Linux is one of popular version of UNIX operating System.
- ▶ It is open source as its source code is freely available.
- ▶ It is free to use.
- ▶ An operating system is a program that acts as an interface between the user and the computer hardware and controls the execution of all kinds of programs.
- ▶ In technical terms, It is a software which manages hardware. An operating System controls the allocation of resources and services such as memory, processors, devices and information.

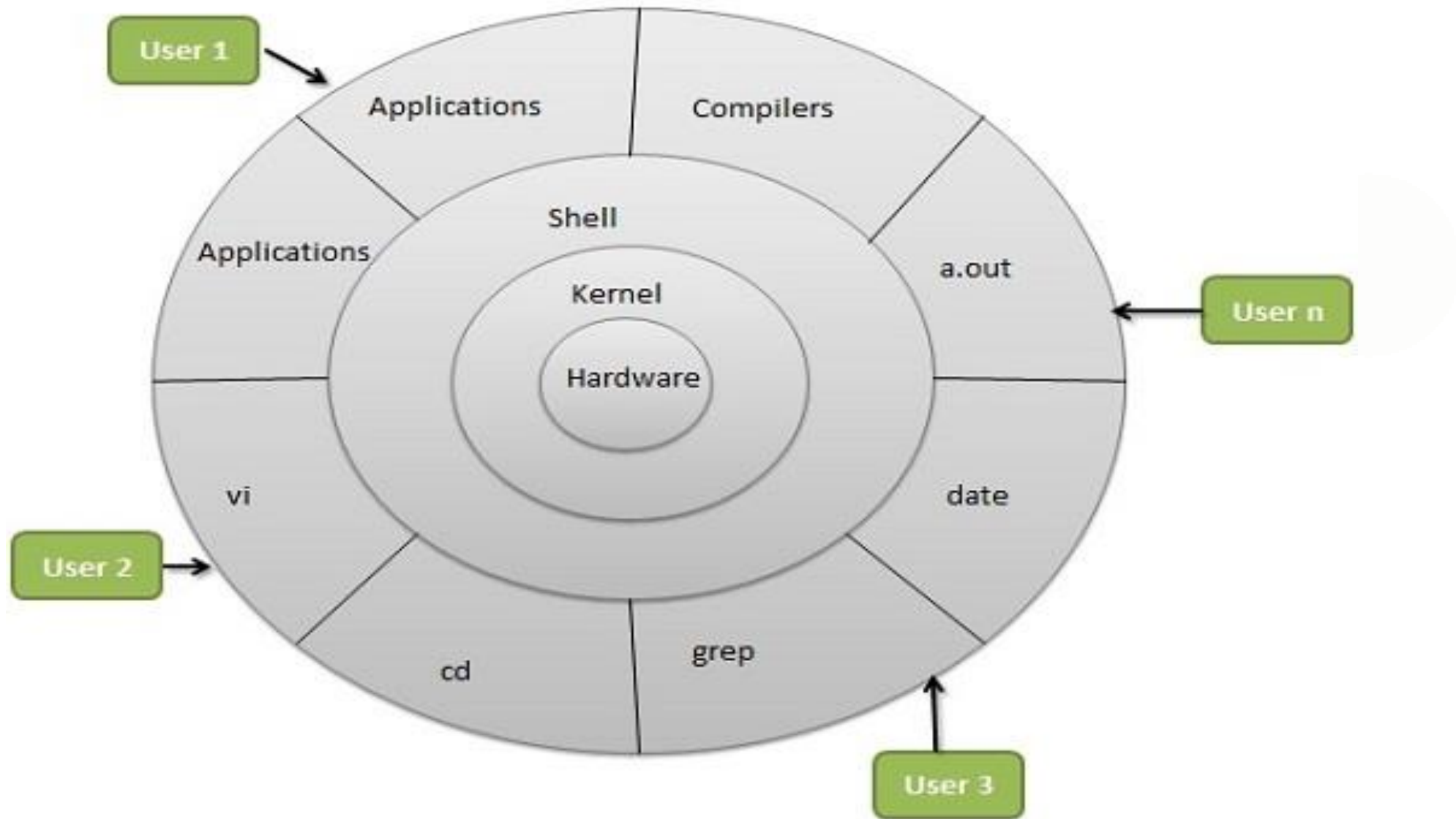
Basic Features

- ▶ Following are some of the important features of Linux Operating System.
- ▶ **Portable** - Portability means software can work on different types of hardware.
- ▶ **Open Source** - Linux source code is freely available and it is a community based development project. Multiple teams work in collaboration to enhance the capability of Linux operating system and it is continuously evolving.
- ▶ **Multi-User** - Linux is a multiuser system means multiple users can access system resources like memory/ ram/ application programs at the same time.

Cont'd

- ▶ **Multiprogramming** - Linux is a multiprogramming system means multiple applications can run at same time.
- ▶ **Hierarchical File System** - Linux provides a standard file structure in which system files/ user files are arranged.
- ▶ **Shell** - Linux provides a special interpreter program which can be used to execute commands of the operating system. It can be used to do various types of operations, call application programs etc.
- ▶ **Security** - Linux provides user security using authentication features like password protection/ controlled access to specific files/ encryption of data.

Linux Architecture



Cont'd

- ▶ Linux System Architecture is consists of following layers
- ▶ **Hardware layer** - Hardware consists of all peripheral devices (RAM/ HDD/ CPU etc).
- ▶ **Kernel** - Core component of Operating System, interacts directly with hardware, provides low level services to upper layer components.
- ▶ **Shell** - An interface to kernel, hiding complexity of kernel's functions from users. Takes commands from user and executes kernel's functions.
- ▶ **Utilities** - Utility programs giving user most of the functionalities of an operating systems.

Linux Basic Commands

What is Command?

- ▶ A **command** is an instruction given by a user telling a computer to do something. **or** Command is nothing but Standard Input, Standard Output and Standard Error.

Basic Commands?

who-- List users logged into the server

#**pwd**-- Show the "present working directory", or current directory.

#**cd**-- Change current directory to your HOME directory.

#**cd /usr/3x3connect**-- Change current directory to /usr/STRIM100.

Cont'd

#cd ..-- Change current directory to the parent directory of the current directory.

#cd.- Current directory

#ls-- list a directory

#ls -l-- list a directory in long (detailed) format

#ls -a-- List the current directory including hidden files. Hidden files start with "."

mkdir <dir name>-- Creates Directory

rmdir <dir name>-- Removes Directory

#more-- Look at file, one page at a time

#touch <file name>-- Creates empty text file with 0 bytes

Cont'd

cat > <file name>-- Create and type file

ctrl + d save & end the file

cat <file name >-- List the contents of the file

#rm -r <dir name>-- Delete directory and its contents

rm <file name>-- Delete the file

#cp file1 file2-- copy a file

cp /tmp/test /var-- Copy tes file to var directory

cp /tmp/test /var/test-- Copy test file to var dir with same name (test)

Cont'd

mv <source> <destination>-- Move file/directory

mv /tmp/test /var-- Move test file to var directory

mv /tmp/test /var/test-- Move test file to var dir with same name (test)

mv /tmp/test /var/test5-- Move test file to var dir and rename it as test5

Recap

- ▶ What is command?
- ▶ How to change the directory?
- ▶ How many ways to create file?
- ▶ How to create directory?
- ▶ How to copy the file in one directory to another?
- ▶ How to move the file in one directory to another?
- ▶ How to remove the file?
- ▶ How to remove the directory?

Vi Editor

vi - Stores 1.48lacs lines or 2gb buffer

vi <file name>-- Create and edit file

esc + i-- insert text

esc + w-- Save

esc + wq-- Save & Quit

esc + dd-- Delete line

esc + u-- Undo

esc + x-- delete character

esc + h-- Move left

esc + l-- Move right

esc + j-- Move down

esc + k-- Move up

esc + o-- Insert line

Cont'd

esc + O-- Insert line above the cursor

esc + w-- Moves word by word

esc + B-- Moves word by word in reverse

esc + yy-- Copy

esc + cc-- Cut

esc + p-- Paste

esc + 100dd-- Delete 100 lines

esc + 100yy-- Copy 100 lines

esc : q-- Quit without save

esc : q!-- Quit without Save forcefully

esc + r-- Replace particular character

esc + cw-- Replace word

esc + : wq!-- Save and Quit

Recap

- ▶ What is the command to go for insert mode?
- ▶ What is the command to go for command mode?
- ▶ How to move the cursor from right to left?
- ▶ How to move the cursor from left to right?
- ▶ How to save and quit the file?
- ▶ How to Replace the word?
- ▶ How to Replace the character?